

Tamanna Subudhi

+1(219)999-7176 | tamanna.subudhi26@gmail.com | [linkedin.com/in/tamanna-subudhi-6792a026a](https://www.linkedin.com/in/tamanna-subudhi-6792a026a)

SUMMARY

Aspiring ML/AI Engineer with hands-on experience designing and deploying end-to-end machine learning pipelines using Python, PyTorch, TensorFlow, and scikit-learn. Skilled in supervised learning, deep learning, computer vision, feature engineering, hyperparameter tuning, and cross-validation, with exposure to MLOps practices including experiment tracking, model versioning, and reproducible model documentation for production-ready systems.

EDUCATION

Purdue University Northwest (PNW) – B.S. Computer Science, AI Concentration | Minor: Mathematics & Statistics | GPA: 3.7/4 | Dean's List *Aug 2022 – May 2026*

TECHNICAL SKILLS

PyTorch, TensorFlow, scikit-learn, Keras, Roboflow, Supervised Learning, Deep Learning, CNNs, Logistic Regression, Random Forest, Gradient Boosting, SVMs, KNN, Neural Networks, Hyperparameter Tuning, L1/L2 Regularization, Ensemble Methods, Feature Selection, Dimensionality Reduction, Cross-Validation, OOD Generalization, Adversarial Robustness, Bayesian Estimation, End-to-End ML Pipelines, Feature Engineering, Data Preprocessing, Normalization, Batch Processing, Experiment Tracking, Model Evaluation (AUC, KS, F1, Accuracy, KS Statistic), ETL Validation, Automated Testing, Model Documentation, Python, SQL, R, Git, GitHub, Jupyter, Pandas, NumPy, Matplotlib, Power BI, Tableau, Excel, JIRA, Agile/Scrum

WORK EXPERIENCE

Undergraduate Researcher – CIVS at Purdue University Northwest

Nov 2025 – Present | Hammond, IN

- Accelerated end-to-end CV model training and evaluation cycles as part of ongoing research and development work, annotating image datasets, building preprocessing pipelines, and fine-tuning CNN-based architectures using Python and Roboflow; iterated on hyperparameter configurations to optimize model convergence.
- Improved experiment reproducibility by maintaining structured model cards documenting preprocessing steps, architecture configs, hyperparameter settings, and evaluation metrics (accuracy, loss curves) across all training runs.

Machine Learning Intern – Poonawalla Fincorp

Jun – Jul 2025 | Mumbai, India

- Engineered features from 50K+ credit records (delinquency ratios, credit utilization, payment history) and built scikit-learn classification pipelines, training Logistic Regression, Random Forest, and Gradient Boosting models with grid search hyperparameter optimization, achieving AUC of 0.87 and KS of 42.
- Operationalized model monitoring by applying product design thinking to build Power BI dashboards to track KPIs, score distributions, and model drift indicators in production, enabling data-driven credit policy decisions by stakeholders.

IT Intern – Sodel Solutions

Jun – Jul 2024 | Navi Mumbai, India

- Reduced ETL pipeline failures by 15% by refactoring SQL queries, implementing automated schema validation, and adding data quality checks across ingestion workflows, improving downstream data reliability.
- Improved data pipeline throughput by 20% by streamlining preprocessing logic, eliminating redundant transformation steps, and standardizing batch processing workflows for faster data load times.

Data Analytics Intern – KPMG

May – Jul 2023 | Mumbai, India

- Reduced manual reporting turnaround by 30% by automating data validation and statistical analysis workflows using Python and SQL, integrating automated anomaly detection and outlier flagging.
- Improved analytics reproducibility and auditability by standardizing KPI definitions, metric documentation, and testing procedures across client consulting engagements using structured reporting templates.

Tutor – PNW Student Academic Support Centre

Aug 2023 – May 2026 | Hammond, IN

- Improved student exam performance by 20% on average through targeted tutoring in Mathematics, Computer Science, and Statistics, adapting problem-solving strategies to individual learning gaps.

PROJECTS

AI Safety – Backdoor Attack Research (BATT)

2025

- Reproduced 'BATT: Backdoor Attack with Transformation-Based Triggers' using PyTorch; implemented rotation(16 degrees) and translation (6px) geometric triggers on CIFAR-10 with a ResNet-18 classifier, achieving ASR of 98.9% and 99.67% respectively at a 5% poisoning rate.
- Benchmarked BATT against the BadNets baseline across benign accuracy and attack success rate; demonstrated that geometric transformation triggers are stealthier and more effective than static pixel patches, with all attacks maintaining benign accuracy within 0.5% of the clean model(76.28%).

ID vs OOD Performance Prediction

2025

- Extended 'Agreement on the Line' paper by replacing its linear ID - OOD assumption with custom nonlinear neural regression models and Bayesian uncertainty estimation on CIFAR-10 and ImageNet, improving OOD performance prediction reliability.
- Demonstrated that uncertainty-aware nonlinear models better capture distribution shift dynamics, contributing insights toward building more generalizable and reliable ML systems.

Heart Disease Risk Prediction

2025

- Trained and benchmarked 6 classifiers (Logistic Regression, Decision Tree, SVM, KNN, Random Forest, Neural Network) with hyperparameter tuning and AUC/F1 evaluation on a multi-state healthcare dataset; performed EDA, feature selection, and cross-validation to identify the best model for early detection.
- Compared model interpretability vs. predictive performance trade-offs across classical ML and neural approaches, selecting optimal architecture for deployment in a healthcare decision-support context.

MNIST Classifier

2025

- Achieved 98% test accuracy on 70K+ images using Logistic Regression with L2 regularization, batch processing, and stratified k-fold cross-validation to maximize generalization on held-out data.
- Applied feature normalization and systematic hyperparameter tuning to optimize model performance, validating results against a held-out test set using accuracy and confusion matrix metrics.

Microsoft Stock Price Prediction – LSTM Neural Network

2025

- Built a stacked LSTM neural network in TensorFlow/Keras to forecast Microsoft closing prices using a 60-day sliding window sequence; applied StandardScaler normalization, Dropout regularization, and Adam optimizer with MSE loss over 20 epochs.
- Performed EDA including open/close trend analysis, volume visualization, and feature correlation heatmap; enforced strict temporal train/ test split (95/5) to prevent data leakage in time series forecasting.

CERTIFICATIONS

Deep Learning Specialization – Coursera/DeepLearning.AI (2025) **Technology Simulation** – Deloitte Australia, Forage (2025)